

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AMRITA SCHOOL OF COMPUTING
AMRITA VISHWA VIDYAPEETHAM – CHENNAI CAMPUS
23CSE302 – COMPUTER NETWORKS

EXPERIMENT 6

ROUTING

Name: Chetan Kumar G		Max Marks (10)
Roll No.: CH.SC.U4CSE24109	Section: CSE B	
Date of Exp: 17-08-2026	Date of Submission: 17-08-2026	

OBJECTIVES:

- 1. Students will gain an indepth experience on the need for routing.**
- 2. Students will understand the types of routing**

A. STATIC ROUTING BETWEEN TWO LAN NETWORKS

AIM

To configure a static router in a network using Cisco Packet Tracer, enabling communication between different network segments by manually setting static routes to direct traffic efficiently.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - 4 PC-PT
- 2) Network Devices
 - 2 Switch 2950-24
 - 2 Routers 2911
- 3) Connections
 - Copper Straight-Through
 - Copper Cross- Over

PROCEDURE

To configure a static router in a network using Cisco Packet Tracer, follow the given steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add four PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag four PCs into the workspace.
- 2) Add two Switches:
 - Click on the "Switches" icon under Network devices in the bottom left corner.
 - Select a switch (2950-24) and drag 2 switches into the workspace.
- 3) Add two Routers:
 - Click on "Routers" icon under Network devices in the bottom left corner.
 - Select a router (2950-24) and drag 2 routers into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PCs to the Switch:
 - Click on a PC and then click on the switch to connect them with a copper straight-through cable, follow the same for another PC.
 - Connect the remaining two PCs with the other switch using the same method.
- 3) Connect the Switch to the Router:
 - Click on the switch and then click on the Router to connect them with a copper straight-through cable (connect Switch0 to Router0's GigabitEthernet0/0 interface).
 - Repeat for the other switch and router (connect Switch1 to Router1's GigabitEthernet0/1 interface).
- 4) Connect the two Routers:
 - Click on a router and then click on the other router to connect them with a copper Cross-Over cable (connect Router0's GigabitEthernet 0/1 interface to Router1's GigabitEthernet0/0 interface).

Step 4: Configure IP Addresses

LAN Network-1 with network address 192.168.1.0/24

- 1) Configure the PC0:
 - Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
 - Choose Static to manually enter the following,

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

- 2) Configure the PC1:
 - Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
 - Choose Static to manually enter the following,

IPv4 Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

3) Configure the Router0:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the switch (GigabitEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

LAN Network-2 with network address 192.168.2.0/24

1) Configure the PC2:

- Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
- Choose Static to manually enter the following,

IPv4 Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

2) Configure the PC3:

- Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
- Choose Static to manually enter the following,

IPv4 Address: 192.168.2.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

3) Configure the Router1:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the PC (GigabitEthernet0/1).
- Set the IP address and subnet mask:

IP Address: 192.168.2.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

WAN Network-2 with network address 10.0.0.0/24

1) Configure Router1:

- Click on the router and navigate to the Config tab.

- Select the interface connected to the Router1 (GigabitEthernet0/1).
- Set the IP address and subnet mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

2) Configure the Router1:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router0 (GigabitEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

Step 5: Configure the Static Routes using either of the 2 methods: Through Command

Line Interface:

(1) Router0

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Configure IP route of the network:
ip route 192.168.2.0 255.255.255.0 10.0.0.2
- 5) Exit configuration mode
End

(2) Router1

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Configure IP route of the network:
ip route 192.168.1.0 255.255.255.0 10.0.0.1
- 5) Exit configuration mode
end

Through Graphical User Interface:

(1) Router0

- 1) Select the router, navigate to Config tab.
- 2) Select Static under Routing and enter the following:

Network: 192.168.2.0

Mask: 255.255.255.0

Next Hop: 10.0.0.2

- 3) Click on add.

(2) Router0

- 1) Select the router, navigate to Config tab.
- 2) Select Static under Routing and enter the following:

Network: 192.168.1.0

Mask: 255.255.255.0

Next Hop: 10.0.0.1

- 3) Click on add.

Step 6: Verify the connectivity between the two LAN Networks:

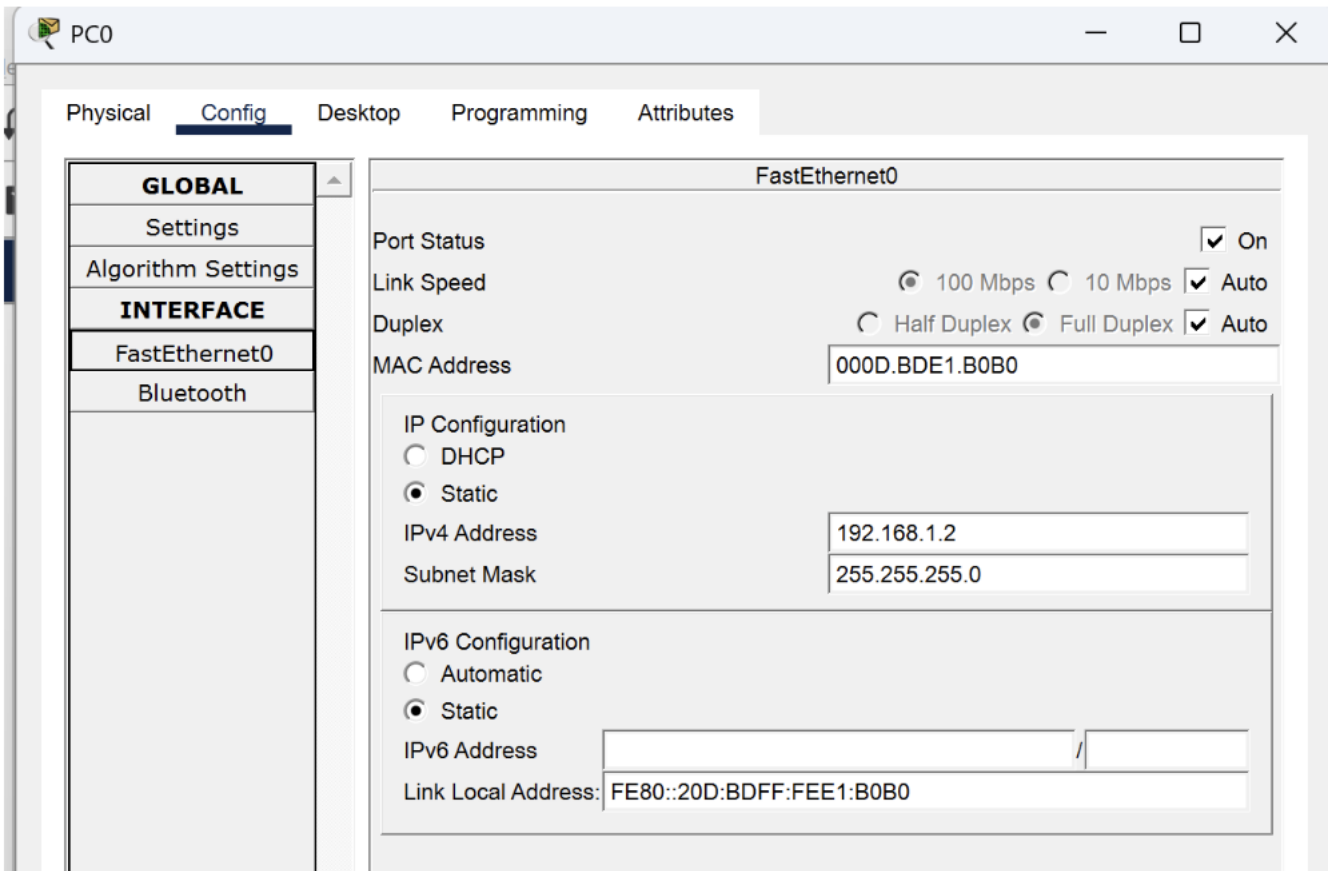
- 1) Ping Test from PC0 to PC2:
 - On PC0, navigate to the Desktop tab and open the Command Prompt.
 - Enter the command **ping 192.168.2.2** and observe the responses.
- 2) Ping Test from PC3 to PC1:
 - On PC1, navigate to the Desktop tab and open the Command Prompt.
 - Enter the command **ping 192.168.1.3** and observe the responses.

NETWORK TOPOLOGY DIAGRAM

OUTPUT

1) Configure IP Address of the PCs

PC0



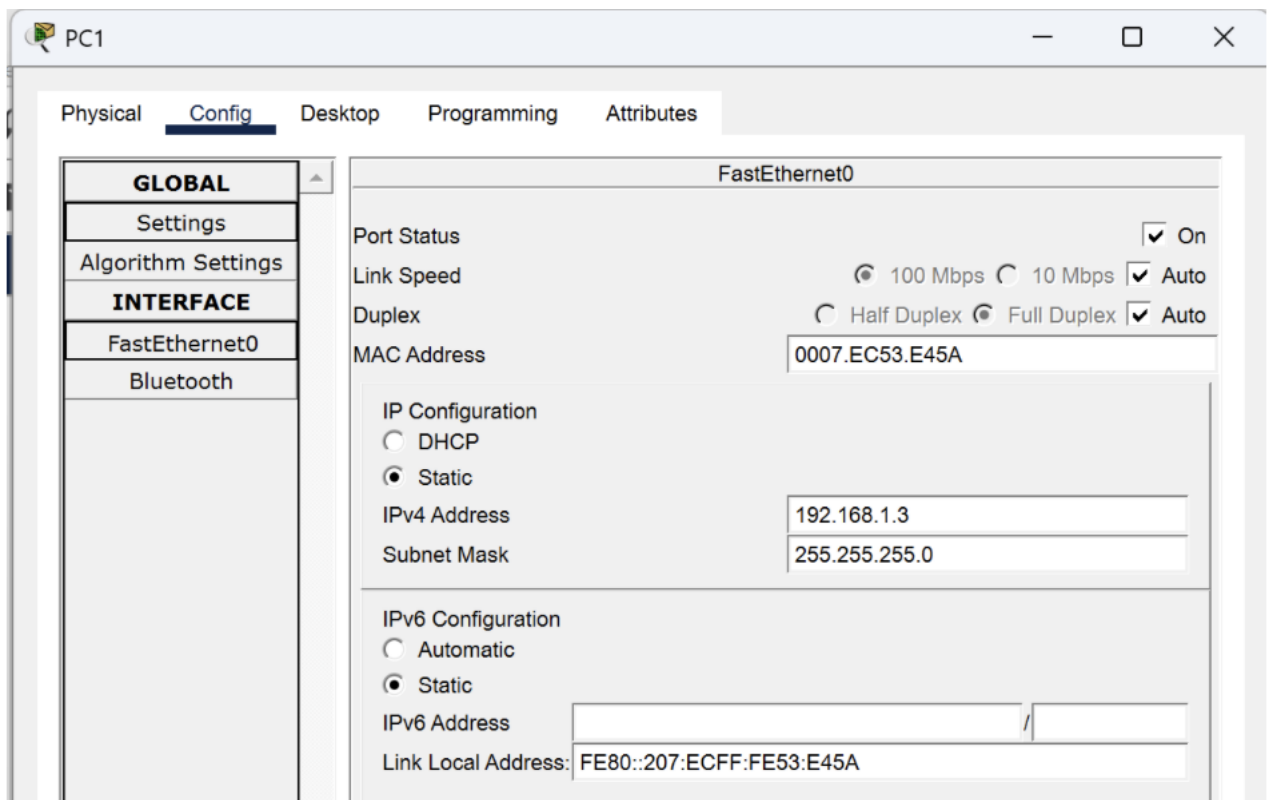
The screenshot shows the configuration window for PC0. The 'Config' tab is selected. On the left, the 'INTERFACE' section is expanded, showing 'FastEthernet0' and 'Bluetooth'. The 'FastEthernet0' configuration is displayed on the right. The 'Port Status' is checked 'On'. 'Link Speed' is set to '100 Mbps' and 'Duplex' is set to 'Full Duplex', both with 'Auto' checked. The 'MAC Address' is '000D.BDE1.B0B0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '192.168.1.2' and the 'Subnet Mask' is '255.255.255.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' field is empty, and the 'Link Local Address' is 'FE80::20D:BDFF:FEE1:B0B0'.

GLOBAL
Settings
Algorithm Settings

INTERFACE
FastEthernet0
Bluetooth

FastEthernet0
Port Status ☒ On
Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address 000D.BDE1.B0B0
IP Configuration
☐ DHCP
☒ Static
IPv4 Address 192.168.1.2
Subnet Mask 255.255.255.0
IPv6 Configuration
☐ Automatic
☒ Static
IPv6 Address
Link Local Address: FE80::20D:BDFF:FEE1:B0B0

PC1



The screenshot shows the configuration window for PC1. The 'Config' tab is selected. On the left, the 'INTERFACE' section is expanded, showing 'FastEthernet0' and 'Bluetooth'. The 'FastEthernet0' configuration is displayed on the right. The 'Port Status' is checked 'On'. 'Link Speed' is set to '100 Mbps' and 'Duplex' is set to 'Full Duplex', both with 'Auto' checked. The 'MAC Address' is '0007.EC53.E45A'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '192.168.1.3' and the 'Subnet Mask' is '255.255.255.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' field is empty, and the 'Link Local Address' is 'FE80::207:ECFF:FE53:E45A'.

GLOBAL
Settings
Algorithm Settings

INTERFACE
FastEthernet0
Bluetooth

FastEthernet0
Port Status ☒ On
Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address 0007.EC53.E45A
IP Configuration
☐ DHCP
☒ Static
IPv4 Address 192.168.1.3
Subnet Mask 255.255.255.0
IPv6 Configuration
☐ Automatic
☒ Static
IPv6 Address
Link Local Address: FE80::207:ECFF:FE53:E45A

PC2

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.F9A9.1D2E

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

PC3

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.3E42.D726

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::260:3EFF:FE42:D726

2) Configure IP Address of Router0

The image displays two screenshots of the Router0 configuration interface, showing the configuration for GigabitEthernet0/0 and GigabitEthernet0/1.

Router0 Configuration - GigabitEthernet0/0

Physical | **Config** | CLI | Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0005.5E63.BD01

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router0 Configuration - GigabitEthernet0/1

Physical | **Config** | CLI | Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0005.5E63.BD02

IP Configuration

IPv4 Address 10.0.0.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

3) Configure IP Address of Router1

The image displays two screenshots of the Router1 configuration interface, showing the configuration for GigabitEthernet0/0 and GigabitEthernet0/1.

Router1 Configuration - GigabitEthernet0/0

Physical	Config	CLI	Attributes
GLOBAL			
Settings			
Algorithm Settings			
ROUTING			
Static			
RIP			
SWITCHING			
VLAN Database			
INTERFACE			
GigabitEthernet0/0			
GigabitEthernet0/1			
GigabitEthernet0/2			

GigabitEthernet0/0

Port Status	☒ On
Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000C.CF4E.4501
IP Configuration	
IPv4 Address	10.0.0.2
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Router1 Configuration - GigabitEthernet0/1

Physical	Config	CLI	Attributes
GLOBAL			
Settings			
Algorithm Settings			
ROUTING			
Static			
RIP			
SWITCHING			
VLAN Database			
INTERFACE			
GigabitEthernet0/0			
GigabitEthernet0/1			
GigabitEthernet0/2			

GigabitEthernet0/1

Port Status	☒ On
Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000C.CF4E.4502
IP Configuration	
IPv4 Address	192.168.2.1
Subnet Mask	255.255.255.0
Tx Ring Limit	10

4) Static Routing Configuration through CLI

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

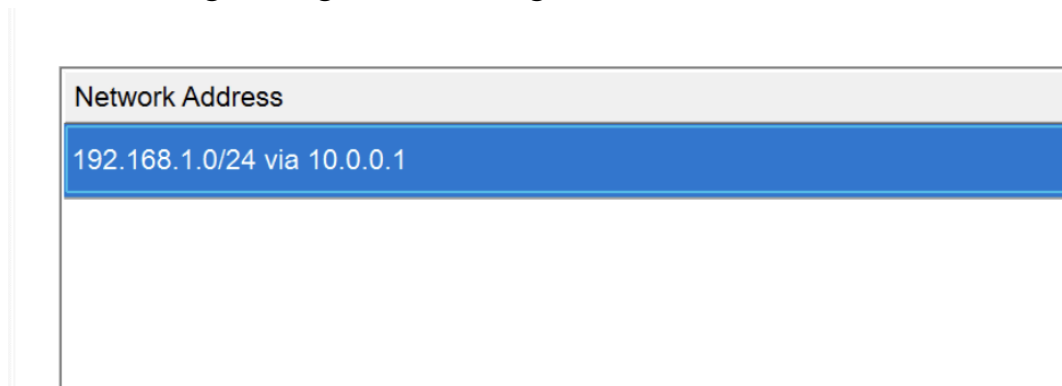
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.2	YES	manual	up	up
GigabitEthernet0/1	192.168.2.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

□

5) Static Routing Configuration through GUI

5) Static Routing Configuration through GUI



6) Verify the connectivity between the two LAN Networks using Ping

6) Verify the connectivity between the two LAN Networks using Ping command

```
Router#ping 192.168.2.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/5 ms
```

```
Router#ping 192.168.1.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
```

```
..!!!
```

```
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms
```

```
□
```

RESULT

After configuring static routes on both routers, PCs in LAN Network 1 should be able to communicate with PCs in LAN Network 2. Successful ping responses between the PCs confirm proper routing and connectivity.

The network connectivity can also be verified by sending packets from PC0 to PC2 in real-time simulation of the Cisco packet tracer,

INFERENCE

Static routing was successfully configured between the two LAN networks. By manually defining routes on both routers, communication between the two different network segments was established. The successful ping responses confirmed that the routers correctly forwarded packets between the LAN networks.

B. DEFAULT ROUTING BETWEEN TWO LAN NETWORKS

AIM

To configure a default router in a network using Cisco Packet Tracer, enabling communication between different network segments by manually setting default routes to direct traffic efficiently.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - 4 PC-PT
- 2) Network Devices
 - 2 Switch 2950-24
 - 2 Routers 2911
- 3) Connections
 - Copper Straight-Through
 - Copper Cross- Over

PROCEDURE:

To configure a default router in a network using Cisco Packet Tracer, follow the given steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add four PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag four PCs into the workspace.
- 2) Add two Switches:
 - Click on the "Switches" icon under Network devices in the bottom left corner.
 - Select a switch (2950-24) and drag 2 switches into the workspace.
- 3) Add two Routers:
 - Click on the "Routers" icon under Network devices in the bottom left corner.
 - Select a router (2950-24) and drag 2 routers into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PCs to the Switch:
 - Click on a PC and then click on the switch to connect them with a copper straight-through cable, follow the same for another PC.
 - Connect the remaining two PCs with the other switch using the same method.
- 3) Connect the Switch to the Router:
 - Click on the switch and then click on the Router to connect them with a copper straight-through cable (connect Switch0 to Router2's GigabitEthernet0/0 interface).
 - Repeat for the other switch and router (connect Switch1 to Router3's GigabitEthernet0/0 interface).
- 4) Connect the two Routers:
 - Click on a router and then click on the other router to connect them with a copper Cross-Over cable (connect Router2's GigabitEthernet 0/1 interface to Router3's GigabitEthernet0/1 interface).

Step 4: Configure IP Addresses

LAN Network-1 with network address 192.168.1.0/24

- 1) Configure the PC0:
 - Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
 - Choose Static to manually enter the following,

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

- 2) Configure the PC1:
 - Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
 - Choose Static to manually enter the following,

IPv4 Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

3) Configure the Router2:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the switch (GigabitEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

LAN Network-2 with network address 192.168.2.0/24

1) Configure the PC2:

- Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
- Choose Static to manually enter the following,

IPv4 Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

2) Configure the PC3:

- Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
- Choose Static to manually enter the following,

IPv4 Address: 192.168.2.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

3) Configure the Router3:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the PC (GigabitEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 192.168.2.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

WAN Network-2 with network address 10.0.0.0/24

1) Configure Router1:

- Click on the router and navigate to the Config tab.

- Select the interface connected to the Router1 (GigabitEthernet0/1).
- Set the IP address and subnet mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

2) Configure the Router1:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router0 (GigabitEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

Step 5: Configure the default Routes using either of the 2 methods: Through Command Line

Interface:

(1) Router2

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Configure IP route of the network:
ip route 0.0.0.0 0.0.0.0 10.0.0.2
- 5) Exit configuration mode
End

(2) Router3

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Configure IP route of the network:
ip route 0.0.0.0 0.0.0.0 10.0.0.1
- 5) Exit configuration mode
end

Through Graphical User Interface:

(1) Router2

- 1) Select the router, navigate to Config tab.
- 2) Select Static under Routing and enter the following:

Network: 0.0.0.0

Mask: 0.0.0.0

Next Hop: 10.0.0.2

- 3) Click on add.

(2) Router3

- 1) Select the router, navigate to Config tab.
- 2) Select Static under Routing and enter the following:

Network: 0.0.0.0

Mask: 0.0.0.0

Next Hop: 10.0.0.1

- 3) Click on add.

Step 6: Verify the connectivity between the two LAN Networks:

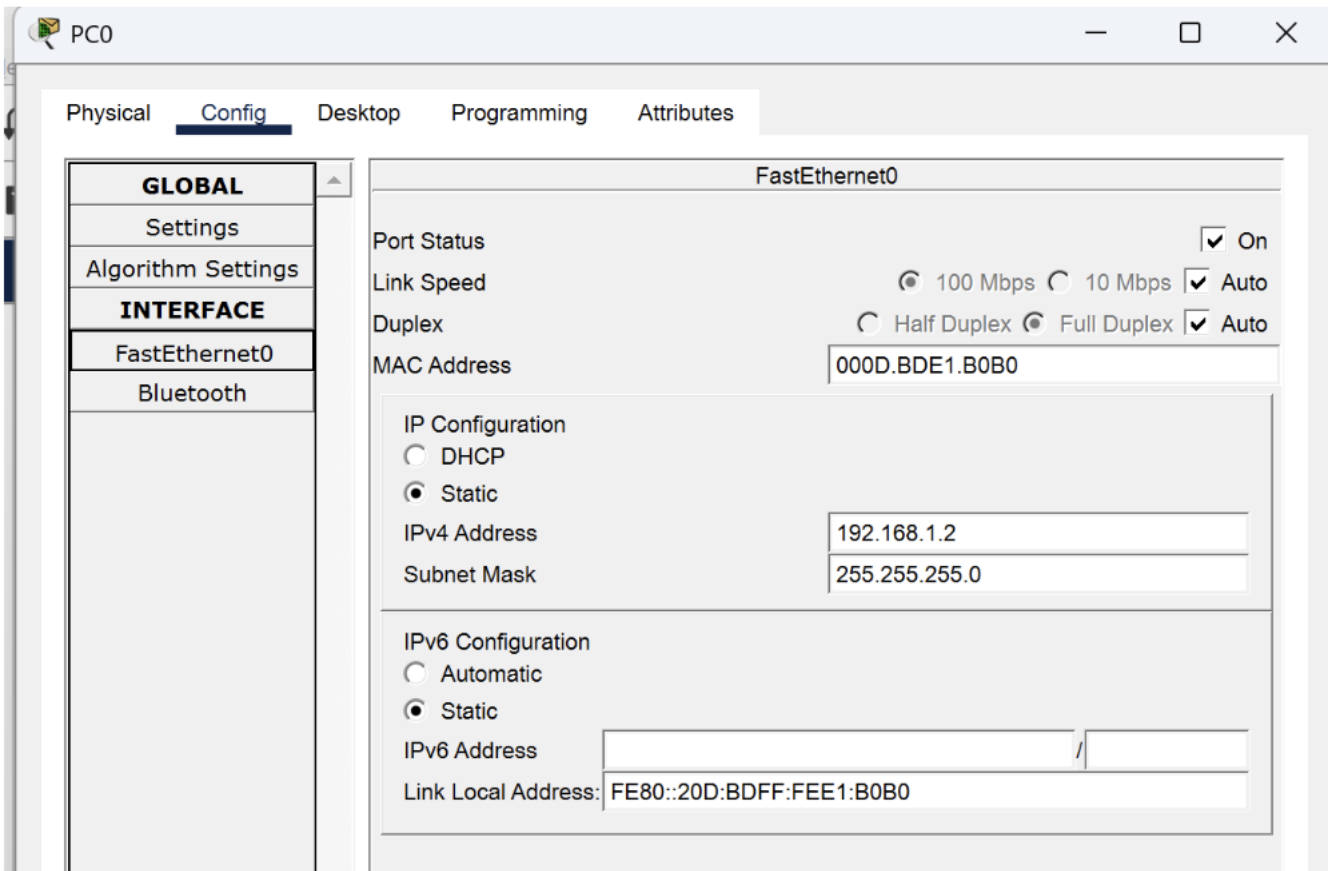
- 1) Ping Test from PC0 to PC2:
 - On PC0, navigate to the Desktop tab and open the Command Prompt.
 - Enter the command **ping 192.168.2.2** and observe the responses.
- 2) Ping Test from PC3 to PC1:
 - On PC1, navigate to the Desktop tab and open the Command Prompt.
 - Enter the command **ping 192.168.1.3** and observe the responses.

NETWORK TOPOLOGY DIAGRAM

OUTPUT

1) Configure IP Address of the PCs

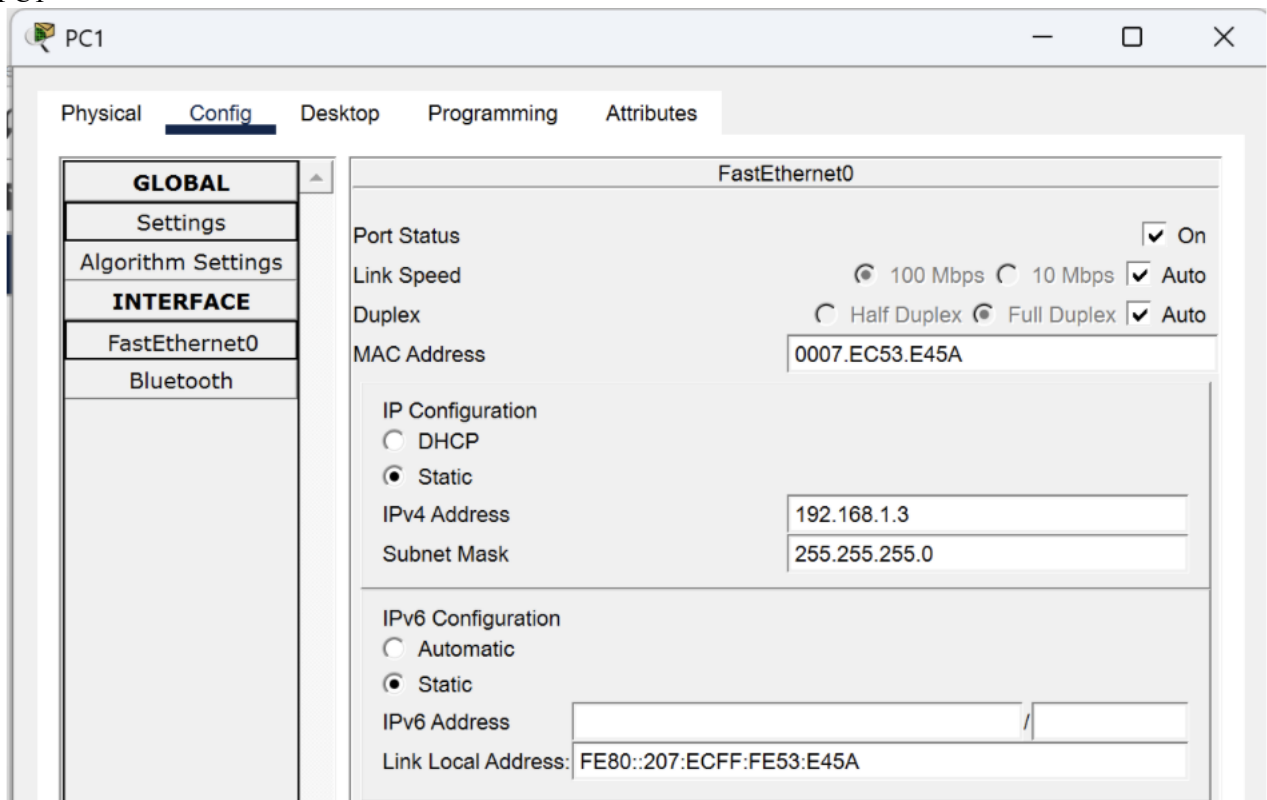
PC0



The screenshot shows the configuration window for PC0. The 'Config' tab is selected. On the left, the 'INTERFACE' section is expanded, showing 'FastEthernet0' and 'Bluetooth'. The 'FastEthernet0' interface is selected. The configuration details for 'FastEthernet0' are as follows:

FastEthernet0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000D.BDE1.B0B0
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
IPv6 Configuration	
☐ Automatic	
☒ Static	
IPv6 Address	
Link Local Address:	FE80::20D:BDFF:FEE1:B0B0

PC1



The screenshot shows the configuration window for PC1. The 'Config' tab is selected. On the left, the 'INTERFACE' section is expanded, showing 'FastEthernet0' and 'Bluetooth'. The 'FastEthernet0' interface is selected. The configuration details for 'FastEthernet0' are as follows:

FastEthernet0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0007.EC53.E45A
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.1.3
Subnet Mask	255.255.255.0
IPv6 Configuration	
☐ Automatic	
☒ Static	
IPv6 Address	
Link Local Address:	FE80::207:ECFF:FE53:E45A

PC2

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.F9A9.1D2E

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

PC3

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.3E42.D726

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::260:3EFF:FE42:D726

2) Configure IP Address of Router0

The image displays two screenshots of the Router0 configuration interface, showing the configuration for GigabitEthernet0/0 and GigabitEthernet0/1.

Router0 Configuration - GigabitEthernet0/0

Category	Setting	Value
Physical	Port Status	☒ On
	Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
	Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
	MAC Address	0005.5E63.BD01
IP Configuration	IPv4 Address	192.168.1.1
	Subnet Mask	255.255.255.0
	Tx Ring Limit	10

Router0 Configuration - GigabitEthernet0/1

Category	Setting	Value
Physical	Port Status	☒ On
	Link Speed	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
	Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
	MAC Address	0005.5E63.BD02
IP Configuration	IPv4 Address	10.0.0.1
	Subnet Mask	255.255.255.0
	Tx Ring Limit	10

3) Configure IP Address of Router1

The image displays two screenshots of the Router1 configuration interface, showing the configuration for GigabitEthernet0/0 and GigabitEthernet0/1.

Router1 Configuration - GigabitEthernet0/0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000C.CF4E.4501

IP Configuration

IPv4 Address 10.0.0.2

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router1 Configuration - GigabitEthernet0/1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000C.CF4E.4502

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

4) Static Routing Configuration through CLI

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

□

5) Static Routing Configuration through GUI

Network Address
0.0.0.0/0 via 10.0.0.2

Network Address
0.0.0.0/0 via 10.0.0.1

6) Verify the connectivity between the two LAN Networks using Ping command

Ping command

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC3

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

RESULT

After configuring static routes on both routers, PCs in LAN Network 1 should be able to communicate with PCs in LAN Network 2. Successful ping responses between the PCs confirm proper routing and connectivity.

The network connectivity can also be verified by sending packets from PC0 to PC2 in real-time simulation of the Cisco packet tracer,

INFERENCE

Default routing was successfully configured between the two LAN networks. By configuring default routes on both routers, packets destined for the other LAN were forwarded through the appropriate next-hop router. Successful ping responses confirmed proper routing and connectivity between the two LAN networks.

C. DYNAMIC ROUTING BETWEEN TWO LAN NETWORKS

AIM

To configure a dynamic router in a network using Cisco Packet Tracer, enabling communication between different network segments by manually setting dynamic routes to direct traffic efficiently.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - 2 PC-PT
- 2) Network Devices
 - 2 Switch 2950-24
 - 3 Router-PT
- 3) Connections
 - Copper Straight-Through
 - Serial DTE

PROCEDURE

To configure a static router in a network using Cisco Packet Tracer, follow the given steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add two PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag two PCs into the workspace.
- 2) Add two Switches:
 - Click on the "Switches" icon under Network devices in the bottom left corner.
 - Select a switch (2950-24) and drag 2 switches into the workspace.
- 3) Add three Routers:
 - Click on the "Routers" icon under Network devices in the bottom left corner.
 - Select a router (Router-PT) and drag 2 routers into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PCs to the Switch:
 - Click on a PC and then click on the switch to connect them with a copper straight-through cable, follow the same for the other PC and switch.
- 3) Connect the Switch to the Router:
 - Click on the switch and then click on the Router to connect them with a copper straight-through cable (connect Switch2 to Router7's FastEthernet0/0 interface).
 - Repeat for the other switch and router (connect Switch3 to Router8's FastEthernet0/0 interface).
- 4) Connect the Routers:
 - Click on Router7 and then click on the Router8 to connect them with a Serial DTE cable (connect Router7's Serial3/0 interface to Router8's Serial3/0 interface).
 - Click on Router7 and then click on the Router9 to connect them with a Serial DTE cable (connect Router7's Serial2/0 interface to Router9's Serial2/0 interface).
 - Click on Router9 and then click on the Router8 to connect them with a Serial DTE cable (connect Router9's Serial3/0 interface to Router8's Serial2/0 interface).

Step 4: Configure IP Addresses

Network-1 with network address 192.168.1.0/24

- 1) Configure the PC:
 - Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
 - Choose Static to manually enter the following,

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

2) Configure the Router7:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the switch (FastEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

Network-2 with network address 155.165.1.0/24

1) Configure the PC:

- Open the PC, navigate to Config, and Edit the FastEthernet0 tab.
- Choose Static to manually enter the following,

IPv4 Address: 155.165.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 155.165.1.1

2) Configure the Router8:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the switch (FastEthernet0/0).
- Set the IP address and subnet mask:

IP Address: 155.165.1.1

Subnet Mask: 255.255.255.0

- Check the box for "Port Status" to bring the interface up.

Network-3 with network address 10.0.0.0/24

1) Configure the Router7:

- Click on the router and navigate to the Config tab. Select the interface
- connected to the Router9 (Serial2/0).Set the IP address and subnet
- mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

Check the box for "Port Status" to bring the interface up.

-

Network-3 with network address 10.0.0.0/24

1) Configure the Router7:

- Click on the router and navigate to the Config tab. Select the interface
- connected to the Router9 (Serial2/0).Set the IP address and subnet
- mask:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

Check the box for "Port Status" to bring the interface up.

-

2) Configure the Router9:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router7 (Serial2/0).
- Set the IP address and subnet mask:

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

Network-4 with network address 30.0.0.0/24

1) Configure the Router9:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router8 (Serial3/0).
- Set the IP address and subnet mask:

IP Address: 30.0.0.1

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

2) Configure the Router8:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router9 (Serial2/0).
- Set the IP address and subnet mask:

IP Address: 30.0.0.2

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

Network-5 with network address 20.0.0.0/24

3) Configure the Router7:

Click on the router and navigate to the Config tab. Select the interface connected to the Router8 (Serial3/0). Set the IP address and subnet mask:

IP Address: 20.0.0.1

Subnet Mask: 255.0.0.0

Check the box for "Port Status" to bring the interface up.

4) Configure the Router8:

- Click on the router and navigate to the Config tab.
- Select the interface connected to the Router7 (Serial3/0).
- Set the IP address and subnet mask:

IP Address: 20.0.0.2

Subnet Mask: 255.0.0.0

- Check the box for "Port Status" to bring the interface up.

Step 5: Configure the Dynamic Routes through Command Line Interface:

(1) Router7

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Start OSPF process with ID 1.
router ospf 1
- 5) Assign network 192.168.1.0/16 to OSPF area 0
network 192.168.1.0 0.0.0.255 area 0
- 6) Assign network 10.0.0.0/8 to OSPF area 0
network 10.0.0.0 0.255.255.255 area 0
- 7) Assign network 20.0.0.0/8 to OSPF area 0.
network 20.0.0.0 0.255.255.255 area 0
- 8) Exit OSPF router configuration mode.
exit

(2) Router9

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Start OSPF process with ID 1.
router ospf 1
- 5) Assign network 10.0.0.0/8 to OSPF area 0
network 10.0.0.0 0.255.255.255 area 0
- 6) Assign network 30.0.0.0/8 to OSPF area 0.
network 30.0.0.0 0.255.255.255 area 0

7) Exit OSPF router configuration mode.

Exit

(3) Router8

1) Select the router, navigate to CLI

2) Enter privileged EXEC mode by typing:

enable

3) Enter global configuration mode by typing:

configure terminal

4) Start OSPF process with ID 1.

router ospf 1

5) Assign network 155.165.1.0/16 to OSPF area 0

network 155.165.1.0 0.0.255.255 area 0

6) Assign network 20.0.0.0/8 to OSPF area 0

network 20.0.0.0 0.255.255.255 area 0

7) Assign network 30.0.0.0/8 to OSPF area 0.

network 30.0.0.0 0.255.255.255 area 0

8) Exit OSPF router configuration mode.

exit

Step 6: Verify the connectivity between the Networks:

1) Ping Test from PC4 to PC5:

- On PC4, navigate to the Desktop tab and open the Command Prompt.
- Enter the command **ping 155.165.1.2** and observe the responses.

NETWORK TOPOLOGY DIAGRAM

OUTPUT

1) Configure IP Address of the PCs

PC0

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0006.2A03.D57C

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::206:2AFF:FE03:D57C

PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.4285.6107

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 155.165.1.2

Subnet Mask 255.255.255.0

IPv6 Configuration

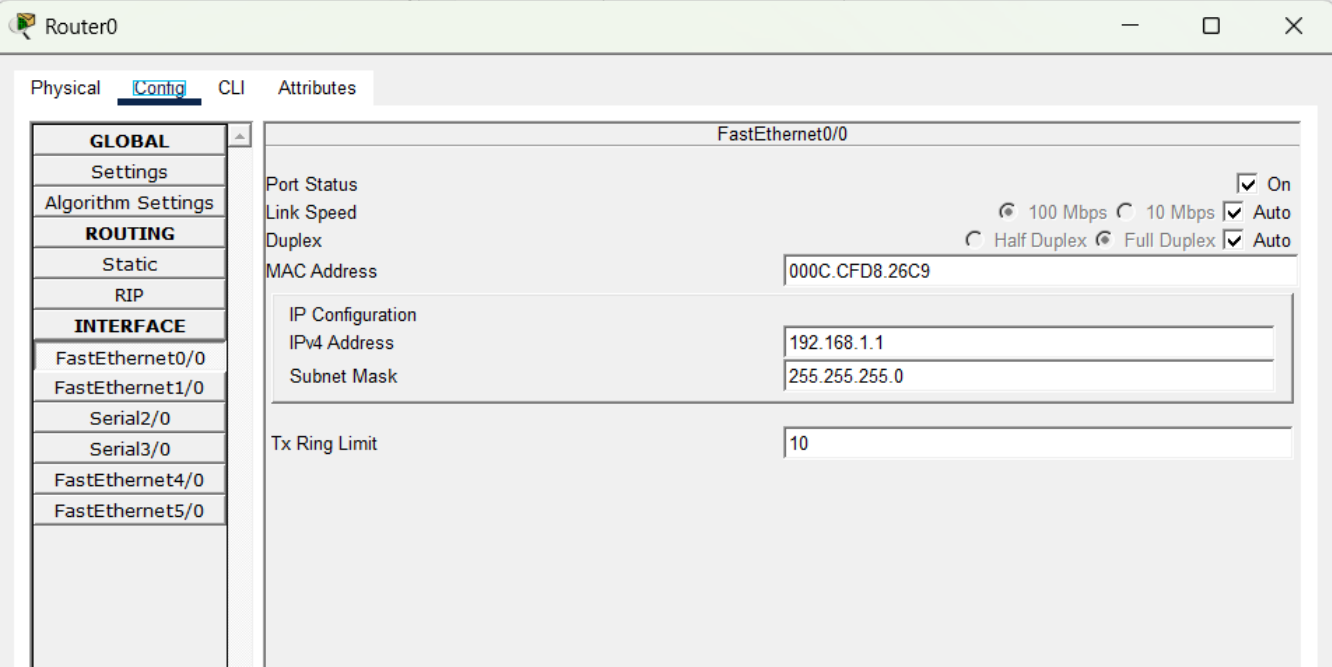
☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::201:42FF:FE85:6107

3) Configure IP Address of Router7



Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- INTERFACE**
- FastEthernet0/0
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

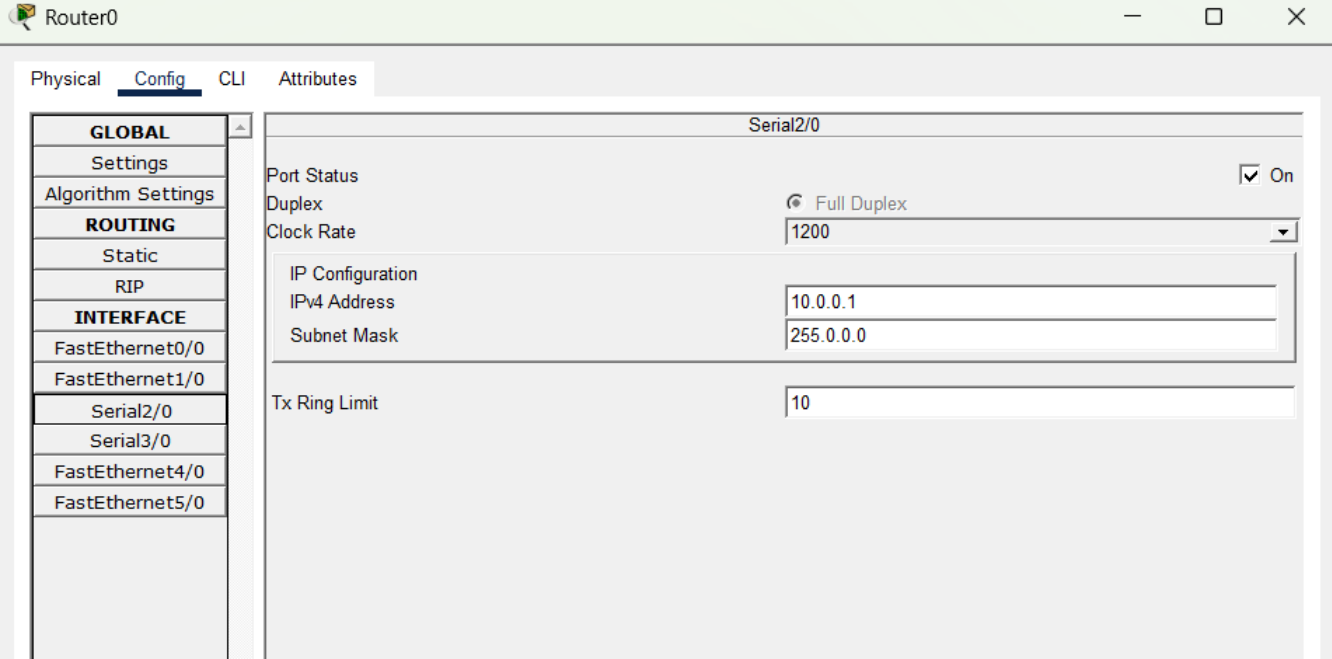
MAC Address 000C.CFD8.26C9

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10



Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- INTERFACE**
- FastEthernet0/0
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 10.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status

On

Duplex

Full Duplex

Clock Rate

1200

IP Configuration

IPv4 Address

20.0.0.1

Subnet Mask

255.0.0.0

Tx Ring Limit

10

4) Configure IP Address of Router8

The image displays two screenshots of the Router1 configuration interface, showing the configuration of two different interfaces.

Top Screenshot: FastEthernet0/0 Configuration

- Physical** | **Config** | CLI | Attributes
- GLOBAL** (selected)
 - Settings
 - Algorithm Settings
- ROUTING**
 - Static
 - RIP
- INTERFACE**
 - FastEthernet0/0 (selected)
 - FastEthernet1/0
 - Serial2/0
 - Serial3/0
 - FastEthernet4/0
 - FastEthernet5/0

FastEthernet0/0 Configuration:

- Port Status: ☒ On
- Link Speed: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address: 00D0.9786.EA2D
- IP Configuration
 - IPv4 Address: 155.165.1.1
 - Subnet Mask: 255.255.0.0
- Tx Ring Limit: 10

Bottom Screenshot: Serial2/0 Configuration

- Physical** | **Config** | CLI | Attributes
- GLOBAL** (selected)
 - Settings
 - Algorithm Settings
- ROUTING**
 - Static
 - RIP
- INTERFACE**
 - FastEthernet0/0
 - FastEthernet1/0
 - Serial2/0 (selected)
 - Serial3/0
 - FastEthernet4/0
 - FastEthernet5/0

Serial2/0 Configuration:

- Port Status: ☒ On
- Duplex: ☒ Full Duplex
- Clock Rate: 2000000
- IP Configuration
 - IPv4 Address: 30.0.0.2
 - Subnet Mask: 255.0.0.0
- Tx Ring Limit: 10

Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status

Duplex

Clock Rate

IP Configuration

IPv4 Address

Subnet Mask

Tx Ring Limit

Full Duplex

On

2000000

20.0.0.2

255.0.0.0

10

5) Configure IP Address of Router9

Router2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☐ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.5C3A.7226

IP Configuration

IPv4 Address

Subnet Mask

Tx Ring Limit 10

Router2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Router2

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status

☒ On

Duplex

Full Duplex

Clock Rate

1200

IP Configuration

IPv4 Address

30.0.0.1

Subnet Mask

255.0.0.0

Tx Ring Limit

10

6) Dynamic Routing Configuration through CLI

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
00:08:37: %OSPF-5-ADJCHG: Process 1, Nbr 30.0.0.1 on Serial2/0 from LOADING to FULL, Loading Done

00:09:56: %OSPF-5-ADJCHG: Process 1, Nbr 155.165.1.1 on Serial3/0 from LOADING to FULL, Loading Done

Router(config)#

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 155.165.1.0 0.0.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
00:09:54: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial3/0 from LOADING to FULL, Loading Done
00:09:56: %OSPF-5-ADJCHG: Process 1, Nbr 30.0.0.1 on Serial2/0 from LOADING to FULL, Loading Done
t

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#
00:08:30: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial2/0 from LOADING to FULL, Loading Done
network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
00:09:51: %OSPF-5-ADJCHG: Process 1, Nbr 155.165.1.1 on Serial3/0 from LOADING to FULL, Loading Done
```

7) Verify the connectivity between the Networks using Ping command

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 155.165.1.2

Pinging 155.165.1.2 with 32 bytes of data:

Reply from 155.165.1.2: bytes=32 time<1ms TTL=128
Reply from 155.165.1.2: bytes=32 time=1ms TTL=128
Reply from 155.165.1.2: bytes=32 time<1ms TTL=128
Reply from 155.165.1.2: bytes=32 time=3ms TTL=128

Ping statistics for 155.165.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

RESULT

After configuring dynamic routes on all three routers, PCs in Network 1 should be able to communicate with PCs in Network 2. Successful ping responses between the PCs confirm proper routing and connectivity.

The network connectivity can also be verified by sending packets from PC4 to PC5 in real-time simulation of the Cisco packet tracer,

INFERENCE

Dynamic routing using OSPF was successfully configured between the three routers. The routers exchanged routing information automatically, established OSPF neighbor relationships, and enabled communication between the two LAN networks. Successful ping responses between PC4 and PC5 confirmed proper connectivity